GENERICS

```
// class
class Box<T> { }
// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) { }
```

JAVA BEING STUPID

```
T obj = new T();            // error
obj instanceof T;           // error
T heh = (T) "asdf";         // no typechecking
var s = Stack<int>();       // no primitive types
private static T x;         // static no workey
Node<Number> s = new Node<Integer>; // error
class IThrowAnErrorBecauseWhyNot {
  void m(List<String> list) { }
  void m(List<Integer> list) { }
} // error because of type erasure & identifiability
```

ITERATOR

```
for (String s : stringList) { } // ==
for (Iterator<String> i = stringList.iterator();
    i.hasNext();) {
  String s = i.next();
}
```

ITERABLE

```
interface Iterable<T> {
  Iterator<T> iterator();
}
interface Iterator<E> {
  boolean hasNext();
  E next();
}
```

COMPARABLE

TODO:
Type signature?

```
static <T extends Comparable> T doStuff(T a, T b) {
  // compareTo is possible with all types
  return a.compareTo("b") > 0 ? a : b;
}
static <T extends Comparable<T>> T doStuff(T a, T b)
{
  // compareTo is possible only with T
  return a.compareTo(b) > 0 ? a : b;
}
```

EXAMPLE

```
class Graphic {
  public void draw() { }
}
class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
    extends Stack<T> {
  public void drawAll() {
    for (T item : this) item.draw();
  }
}
```

BYTECODE

TODO:

TYPE ERASURE

- Introduced because of "bAcKwArDs CoMpAtAbIlItY"
- No type information at runtime (Non-Reifiable Type)

```
class MyStack<T> {
  Entry<T> top;
  int push(T value) { }
}
// becomes
class MyStack {
  Entry top;
  int push(Comparable value) { }
}
/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();
```

WILDCARDS

```
void printGs(List<Graphic> gs) { }
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1);   // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2);   // error
/* solution */
void printGs(List<? extends Graphic> gs) { }
```

GENERIC VARIANCE

	Type	Compatible Type-Args	R	W
Invariance	C<T>	T	✓	✓
Covariance	C<? extends T>	T and sub-types	✓	✗
Contravariance	C<? super T>	T and basis-types	✗	✓
Bivariance	C<?>	All	✗	✗

INVARIANCE

```
static <T> void move(Stack<T>, from, Stack<T> to) {
  while(!from.isEmpty()) to.push(from.pop());
}
var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle());          // ok
move(rectS, graphS);                   // error
Stack<Object> objS = graphS;           // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA;                  // ok
objA[0] = Integer.valueOf(2);          // runtime err
```

TODO:

```
// compiles?
List<Graphic> list = new
ArrayList<Rectangle>();
// compiles?
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new
Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new
Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
for (Rectangle r : rectangleStack) {
  graphicStack.push(r);
}
```

COVARIANCE

```
public class Stack<E> {
  public void pushAll(Iterable<? extends E> src) { }
}
Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>();     // ok
graphS = new Stack<Circle>();        // ok
graphS = new Stack<Object>();        // error (duh)
/* read only */
Stack<? extends Graphic> stack = new
  Stack<Rectangle>();
stack.push(new Graphic());           // error
stack.push(new Rectangle());         // error
stack.push(new Triangle());          // error
```

CONTRAVARIANCE

```
static <T> void addToC(List<? super T> list, T e) {
    list.add(e);
}
addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok
/* shenanigans */
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic());                // ok
s.add(new Rectangle());              // ok
Graphic g = s.pop();                 // error
Object g = s.pop();                  // ok
```

BIVARIANCE

```
static void printList(List<?> list) {
  for (Object elem : list)
    System.out.print(elem + " ");
}
/* more shenanigans */
static void appendNewObject(List<?> list) {
  list.add(new Object());            // error
}
```

PRODUCER / CONSUMER

TODO:

MISC

```
<T extends Comparable & Collection> // multiple type
bounds
```

GENERICS STREAM TOMFOOLERY

```
List<? extends Media> mediaList;
public <S extends T> List<S>
  filterBySubtype(Class<S> subtype) {
    return mediaList.stream()
            .filter(subtype::isInstance)
            .map(subtype::cast)
            .collect(Collectors.toList());
}
```

ANNOTATIONS

- Metadata
- Not actually part of the code

TODO:
slides 10

- @Override
- @Test
- @Json...

DEFINING

```
@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
  String name();
  String date() default "N/A";
}

@Author(name = "UwU", date = "idon'tvalidateinput")
public class WeeOoo { }
```

REFLECTION

- Information of Class (private and public): Methods, Attributes, Parent class, Implemented interfaces
- Calling methods possible
- Usages: Debugger, Serialization, Frameworks, Remote Procedure Call, ORM

```
// all of these `throws SecurityException`
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<?>[] getDeclaredConstructors()

System.out.println(dump(new Student("a", "b"), 0));
// {
//   firstName = a
//   lastName = b
// }
Class c = "s".getClass(); // java.lang.String
```

TODO:
Diagram (slides 26)

METHOD

```
// @Target(ElementType.METHOD)
public String getName()
public boolean isAnnotationPresent(
  Class<? extends Annotation> annotationClass)
public Object invoke(Object obj, Object... args)
  throws IllegalAccessException,
         IllegalArgumentException,
         InvocationTargetException

public class Profiler {
  public static void main(String[] args) {
    var testFunctions = new ProfileFunctions();
    int[] array = {5, 2, 8, 1, 93, 3, 33, 1, 333};
    var methods = ProfileFunctions.class
                    .getDeclaredMethods();
    for (var m : methods) {
      if(m.isAnnotationPresent(Profile.class)) {
        m.setAccessible(true); // best practice or
sth, idk, i write code in actually good languages
        Profiler.profileMethod(testFunctions, m,
          new Object[] {array});
      }
    }
  }
}
```

CLASS

TODO:

```
// @Target(ElementType.TYPE)

getDeclaredConstructor().newInstance()
...
```

ATTRIBUTE

TODO:

```
// @Target(ElementType.FIELD)

...
```

VALIDATION

```
@Min(value = 18, message = "Age must be ⩾ 18")
@Max(value = 99, message = "Age must be ⩽ 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;
```

ALGORITHMS

- Brute-force
- Greedy
  - Take best option each step (optimize locally)
  - Fast
- Backtracking
  - Trial and error
  - Best overall option
  - Higher compute costs
- Divide and conquer
  - Binary search
- Dynamic programming
  - Recursion optimization?
  - Tabulation

BACKTRACKING

TODO:
slides 87

BIG O NOTATION

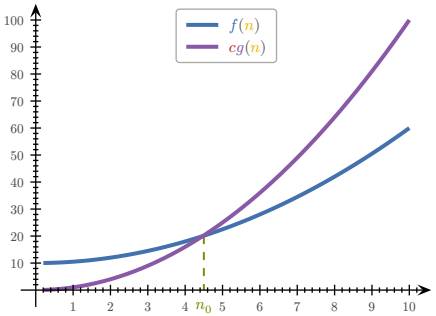
- Worst case scenario
- Atomic operations = constant time
- Runtime measured as sum of primitive operations

Notation

$$\overset{\text{Algorithm}}{\widehat{f}}\left(\underbrace{n}_{\text{Size of problem}}\right) = \text{Number of steps}$$

g = complexity class

TODO:
diagram $f \leq cg$



$f(n)$ is $O(g(n))$ if $c > 0 \wedge n_0 \geq 1$

$f(n) \leq cg(n)$ for $n \geq n_0$

f in order g:

$$\text{Let } f, g : \mathbb{N} \to \mathbb{N}$$
$$n_0, c \in \mathbb{N} \wedge \forall n \geq n_0 : f(n) \leq c \cdot g(n)$$
$$\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$$

Big O

$$\overset{\text{Big O}}{\widehat{O}}\left(\underbrace{n}_{\text{Size of problem}}\right) = \text{Complexity class}$$

RULES

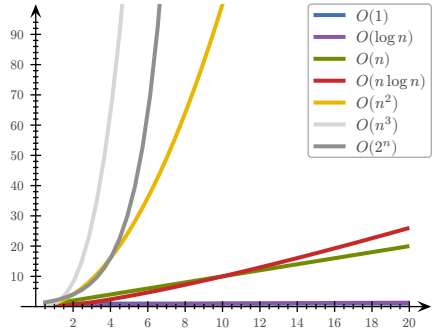
If $f(n)$ is polynomial of degree d, then $f(n) \in O(n^d)$
- ignore lower powers
- ignore constants

Use most optimal function (lowest power)
- $2n$ is $O(n)$ better than $2n$ is $O(n^2)$

Simplify as far as possible
- $3n + 5$ is $O(n)$ instead of $3n + 5$ is $O(3n)$

COMPLEXITY CLASSES

Name	Class	Example
Constant	1	Array read
Logarithmic	$\log n$	Binary search
Linear	n	Linear search
Log-linear	$n \log n$	Merge+Quick sort
Quadratic	n^2	Bubble+Insertion sort
Cubic	n^3	Solve equation
Polynomial	n^k	Various algorithms
Exponential	2^n	Calculate all subsets
Factorial	$n!$	Calculate all permutations



TODO:
Counting primitive operations (slides 55)

SORTING ALGORITHMS

SELECTION SORT

- Search for smallest elem in unsorted part and move in front

– Less mutations in array
– Same performance even if list almost sorted

```
public static void selectionSort(int[] a) {
    int n = a.length;
    for (int i = 0; i < n - 1; i++) {
        int minimum = i;
        for (int j = i + 1; j < n; j++)
            if (a[j] < a[minimum])
                minimum = j;
        swap(a, i, minimum);
    }
}
```

$$(n-1)+(n-2)+\ldots+1=\sum_{i=1}^{n-1} i=\frac{n(n-1)}{2}=\frac{n^2-n}{2}\sim O(n^2)$$

INSERTION SORT

– Take elem and put into correct place in sorted part
– Good for already partially sorted arrays

```
public static void insertionSort(int[] a) {
  int n = a.length;
  for (int i = 1; i < n; i++)
    for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
      swap(a, j, j - 1);
}
```

$$1+\ldots+(n-1)+n=\sum_{i=1}^{n} i=\frac{n(n+1)}{2}=\frac{n^2+n}{2}\sim O(n^2)$$

BUBBLE SORT

– Compare neighboring elems and swap if needed

```
public static void bubbleSort(int[] a) {
  for (n = a.length; n > 1; n--)
    for (i = 0; i < n - 1; i++)
      if (a[i] > a[i + 1])
        swap(a, i, i + 1);
}
```

$$(n-1)+(n-2)+\ldots+1=\sum_{i=1}^{n-1} i=\frac{n(n-1)}{2}=\frac{n^2-n}{2}\sim O(n^2)$$

COUNTING SORT

```
public static void countingSort(int[] a) {
  int k = Arrays.stream(arr).max().getAsInt();
  int[] c = new int[k + 1];
  for (int i : a) c[i] += 1;
  int pos = 0;
  for (int i = 0; i < k; i++)
    for (int j = 0; j < c[i]; j++) {
      a[pos] = i;
      pos++;
    }
}
```

$$n+n+k+n=3n+k\sim O(n+k)$$

SHELL SORT

– Sort pairs of far away elems
– Sort the partially sorted array through insertion sort

```
public static void shellSort(int[] a) {
  int n = a.length;
  int h = 1;
  while (h < n/3) h = 3 * h + 1;
  while (h ⩾ 1) {
    for (int i = h; i < n; i++)
      for (int j = i; j ⩾ h && a[j] < a[j - h];
           j = j - h) swap(a, j, j - h);
    h /= 3;
  }
}
```

BINARY SEARCH

```
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
  if (low > high) return false;
  int pivot = low + ((high - low) / 2);
  if (target.equals(data.get(pivot)))
    return true;
  else if (target.compareTo(data.get(pivot)) < 0)
    return searchBinary(data, target, low,
                        pivot - 1);
  else
    return searchBinary(data, target, pivot + 1,
                        high);
}
```

$$O(\log n)$$

RECURSION

When a function calls itself

LINEAR RECURSION

Recursive call starts ***at most one*** further recursive call

> ***TODO:***
> recursive → iterative (slides 61)

TAIL RECURSION

Linear recursion where the recursive call is ***last***

> ***TODO:***
> recursive → iterative (slides 69 – nice)

BINARY RECURSION

All non-terminal calls have ***two*** recursive calls

> ***TODO:***
> example (slides 76,77,78)

> ***TODO:***
> $O(n)$ (slides 84)